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Harnessing Digital Transformation for Sustainable Performance: Exploring the Mediating Roles of Green Knowledge Acquisition and Innovation Performance Under Digital Transformational Leadership

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Abstract: The rapidly growing proliferation of digital technology presents unprecedented opportunities to address sustainability challenges in today's manufacturing sector. This research examines the effects of digital transformation on sustainable corporate performance, focusing on the mediating roles of green knowledge acquisition and innovative performance. Grounded in new institutional theory, this research targets the Turkish manufacturing industry, employing a quantitative methodology. Data were collected via structured surveys from 332 middle and senior managers, representing diverse sectors, using purposive sampling. Statistical analyses, including regression and moderated multiple regression, were executed using AMOS and Hayes' PROCESS macro. Findings demonstrate that digital transformation positively influences sustainable corporate performance. Green knowledge acquisition and innovation performance serve as significant mediators. Moreover, digital transformational leadership was found to enhance the relations between digital transformation and its mediators. However, limited leadership capabilities weakened these connections. This result highlights the significance of digital tactics with sustainable goals to achieve competitiveness with eco-friendly corporate practices. These findings significantly enhance sustainable performance by facilitating the acquisition of green knowledge and fostering innovation performance. The implications emphasize the necessity for policymakers and business leaders to foster digital leadership, enhance innovation capabilities, and promote green knowledge management as part of their organizational strategies. The research study investigating organizations can leverage digital technologies to achieve sustainable performance by actively acquiring green knowledge and driving innovation. This research introduces a novel distinction into the mechanism through digital transformation that can drive sustainability, thus contributing to a duality of academic literature and practical implementation in a period of rising ecological and economic difficulties.

Keywords: digital transformation; Turkish manufacturing industry; sustainable corporate performance; green knowledge acquisition; innovation performance; digital transformational leadership; new institutional theory



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1. Introduction

In recent years, the impact of global warming has gathered increasing importance as a global concern, sparking extensive debate and increasing awareness in academic literature [1,2]. Despite this heightened attention, human activities continue to degrade

ecosystems, exacerbating the environmental challenges. The scientific community broadly agrees that climate change is primarily driven by anthropogenic factors, including industrial activities, particularly by the manufacturing sector. These activities are a major contributor to rising carbon emissions, which significantly expand the global carbon footprint [3].

Despite these difficulties, international efforts such as the European Union's Eco-innovation Action Plan, the 2021 Glasgow Climate Agreement, and the 2016 Paris Agreement highlight the global consensus on addressing sustainable development goals. These initiatives focus on critical objectives such as reducing carbon emissions, promoting renewable energy adoption, and conserving the natural ecological environment [4,5]. Continually, organizations are increasingly being compelled to adapt their operational and strategic frameworks to integrate green knowledge and advance eco-innovation practices. Such efforts are essential for achieving the overarching targets of sustainable development and reducing the adverse environmental consequences [6].

Advances in digital technologies enable firms to adapt production processes and design and produce eco-friendly products to address environmental conservation concerns. Based on this, digital transformation becomes a critical catalyst for manufacturers to promote sustainability. This offers alternative ways to enable decision-making, optimize operations for resource efficiency, foster innovation in green technologies, improve supply chain transparency, and empower customers to make sustainable choices. By definition, digital transformation involves the extensive overhaul of a firm's operations, product design, development, manufacturing, provision of services, or sale of their products [7]. The integration of digital technology is important for increasing the number of business activities to support innovation and sustainable growth because digital transformation enables firms to effectively acquire and portray information and the flow of data and make improved decisions, enabling resource efficiency, reducing garbage and pollution control [8]. While this might seem straightforward, actualizing this capacity poses an advanced challenge for the firms that must leverage digital transformation to derive valuable data that can aid decision-making and achieving sustainable corporate performance [9–11].

Agile and adaptive manufacturers are striving to build organizational capabilities to guide knowledge management processes within their firms [11,12] to allow them to accomplish sustainable corporate performance. From this standpoint, information has become an important element that allows firms to succeed and achieve sustained performance [13]. Suitable knowledge management helps select the most relevant strategies that enhance the efficacy of decision-making procedures that favorably influence the firm's outcomes [13,14]. Gaining, distributing, and using knowledge regarding environmental preservation can assist firms in accomplishing market dominance [8,15]. Green knowledge acquisition is a common process of actively gaining knowledge about eco-friendly practices and technologies. Correspondingly, the perception of green knowledge acquisition driven by the firm's capacity to digitalize allows organizations to adapt quickly to market development and make their operations more sustainable, boosting stakeholder and consumer confidence and improving corporate image [11,16]. However, research investigating the relationship between digitalization with green knowledge acquisition is still in its infancy, with the current efforts in this research area often constrained by the investigation scope. For instance, some researchers demonstrated that digital capability promotes green knowledge creation [5]; despite this, the current literature lacks empirical studies that address the influences of digitalization on other measures of green knowledge like green knowledge acquisition. Green knowledge acquisition relates to the activities of the organizations to acquire knowledge relating to ecological security [17]. Overall, it leads to innovative and impactful solutions that contribute to sustainable development, often involving new approaches and outcomes that go beyond traditional sustainability methods.

Given global competitiveness, emerging research has highlighted that digital transformation can contribute to economic efficiency and organizations' innovation fulfillment [18–20]. Despite their importance, these studies were conducted outside the scope of sustainability. Moreover, little is known about how digital transformation can influence innovation performance for sustainable corporate performance. This study's objectives are used to analyze the relationship between digital transformation and innovation performance, as they can be essential ingredients for achieving sustainable corporate performance.

While the current literature highlights digital transformation's importance on performance-related outcomes, scholars have different views of the relationship. For example, some suggest that data analytics platforms can promote resource coordination and internal and external communication, ultimately boosting performance. Further, considering that digital transformation requires huge funding in terms of both human resources and finance and firms also have to re-evaluate and remake their firm's structure and operational procedures [21], it may influence performance unfavorably if a firm does not have sufficient resources or is unsuccessful in reconfiguring the structure and processes necessary for digital transformation. These attempts suggest that digital transformation on sustainability remains under-researched [2].

Although numerous studies examine the gap in the literature [17,22,23], previous study investigated how effective leadership in the digital age can drive environmentally conscious innovation through the acquisition of relevant green knowledge. Some researchers explained the direct relationship between green knowledge acquisition and sustainable corporate performance in multinational manufacturing [24]; others demonstrate that knowledge acquisition has no impact on the sustainability of service and manufacturing business. It is explained by managers and leaders of these companies not adequately acknowledging the value of knowledge. Thus, this study proposes a discussion of the interaction of knowledge acquisition by incorporating transformational leadership into the framework [25].

In the previous research, concerning small and medium enterprise owners and senior managers navigating sustainable digital transformation challenges, it was critical to bridge the information gap by providing targeted resources and guidance specifically tailored to their needs [26]. To overcome this gap in the literature, the study embraces sustainability performance, green knowledge acquisition, and innovative performance and effectively leverages digital technology leadership in the manufacturing sectors [27]. The recent research of Xu et al. [2] highlights the positive role of digital transformation on sustainable performance. However, little is known about how (i.e., the mechanisms) this relationship develops. To fill this gap, this research will examine the mediating role of green knowledge acquisition and innovation performance.

Furthermore, with the profound and disruptive changes brought by digital transformation, leadership must adapt in various ways concurrently [26,28]. Moreover, management plays a critical part in inspiring stakeholders and establishing a foundation for taking action [29]. Leaders who embrace digital transformation are often referred to as "digital leaders"; they associate collaborative networks and maintain expertise in the digital domain [30]. Although the literature review on digital transformation specializes in the importance of transformational leadership, there is, however, limited literature available on digital transformation leadership despite the role leadership plays in digital transformation [26,31]. Since leadership plays a critical part in digital transformation, drawing from dynamic capabilities theory [32] and new institutional theory [24], this study investigates the impact of digital transformation on sustainable corporate performance through the mediating role of green knowledge acquisition and innovation performance and the moderating role of digital transformational leadership. This digital transformational leadership can be assessed through a combination of behavioral observation, performance metrics,

and leadership development programs. These practices moderate the relationship between green knowledge acquisition and innovation performance in the Turkish manufacturing industry. Taken together, this research addresses the following questions, building on the discussed research gaps in the literature.

RQ1: Does the digital transformation influence green knowledge acquisition, innovation performance, and sustainable corporate performance?

RQ2: Is the relationship between digital transformation leadership and sustainable corporate performance mediated by green knowledge acquisition and innovation performance?

RQ3: Does digital transformational leadership further improve the relationship between digital transformation, green knowledge acquisition, innovation performance, and sustainable corporate performance?

This empirical research aims to offer new theoretical contributions to the literature on digital transformation and sustainability by applying the new institutional theory as foundational theory, investigating the impact of digital transformation on sustainable corporate performance in the manufacturing industry, and simultaneously underlining the meaningful mediating roles of green knowledge acquisition and innovation performance in this relationship in an integrated parsimony framework. The purpose of this study is to understand how implementing digital technologies within an organization can drive sustainable practices by facilitating the acquisition of environmentally conscious knowledge and fostering innovative solutions that minimize environmental impact, all facilitated by leadership focused on digital transformation strategies. Thus, this research bridges the gap concerning digital transformation to expand sustainable corporate performance, especially when the current literature specifies the significance of digital transformation for sustained development [8,33]. Moreover, the emerging literature demonstrates that digital transformation significantly influences sustainable corporate performance [2]; despite this, few are clear about the conditions under which this relationship develops. Specifically, not much is known in the current literature about the managerial and organizational conditions that could make this relationship more or less pronounced. As part of this work, we aim to uncover digital transformation leadership as a driving catalyst that can inhibit or enhance the relationships in our integrated parsimony framework. The findings of this research have the potential to offer crucial insights to practitioners in the manufacturing industry that can help clarify the relevance of the roles of green knowledge acquisition and innovation performance in improving firms' sustainable corporate performance while also guiding these firms in their operational strategies in preparing for the challenges of the digitalization era and sustainability.

2. Theoretical Background and Hypotheses Development

2.1. The New Institutional Theory (NIT)

This study is motivated by a well-established theory: the new institutional theory. The new institutional theory is frequently adopted to explain firm changes such as processes, new rules, and as well as overall organizational structures. More recently, the digital transformation literature has adopted the theory to investigate different aspects of the process [34,35]. Further, it has also been used as a guiding principle to investigate external factors affecting organizational cultures and practices regarding technology development [2,34]. Some external factors that affect organizational cultures are the norms and standards, government regulations, competitor behavior, market demands, and broader technological trends. These exert pressure on organizations to adopt new technologies and adapt their practices to remain legitimate and competitive within their field [36].

Furthermore, the new institutional theory has been used to explain innovation and transformation from a socio-cultural standpoint via two separate research analyses. First, it investigates the connection between stability and change, considerations of the variability and heterogeneity of firms as well as their continuity and homogeneity [37]. Second, consistency and modification can be analyzed from various perspectives, including various structures, activities, organizations, and individuals [38]. Sustainability warrants organizations adjusting their operational strategies for environmental preservation using digital technologies [8]. However, limited knowledge exists in the current literature regarding the application of the new institutional theory to investigate digital transformation and sustainability. Therefore, this research effort uses the new institutional theory as a guiding principle to investigate how digital transformation translates into sustainable corporate performance to connect practice and theory. The theory will also be used to explain how actors such as those in leadership roles may add to digital transformation to achieve sustainable development targets.

2.2. Digital Transformation

Digital transformation involves intentional changes leveraged upon advanced digital technology [30]. Digital transformation relates the structural changes to using big data analytics and other correspondence technology to offer products and services [30,39]. It was also described as a tool for changing business operations, cultures, and organizational aspects to adapt to changes in market requirements instigated by digital technologies [40]. It has many other transformations, depending on an organization's ability to develop new requisite capabilities [41]. The trio of hallmarks of digital transformation: (a) re-examining as well as redefining firm borders, (b) offering community response and diminishing property rights, and (c) reshaping the identity of products and the firm. From the new institution theory perspective, digital transformation would profoundly transform firms and their associated fields [42].

In this research, digital transformation relates to the combination of different digital technologies and digital innovations, leading to the emergence of new practices, frameworks, structures, beliefs, values, new actors, and structures that disrupt and amend already established ecosystems, fields, and industries. From this standpoint, Scuotto et al. [43] stated that these adjustments influence the recruitment of employees and leaders while also revamping the culture of the firm.

Technologies like search engines, web analytics, and social media promote digital communication between firms and their stakeholders and develop reliable and effective digital capabilities [2,44,45]. A firm must integrate digital technologies into its processes, interiorly and exteriorly; thus, the adoption of digital transformation is necessary to build on digital tactics. Due to the ever-changing, complex nature of the digitalization process, businesses must continuously develop, execute, assess, and adapt their preferences to change their operations. Digital transformation becomes imperative to delve into the historical underpinnings and the transformative journey that has brought us to the present digital landscape [26,46]. Internal diffusion of digital technologies and integration can only be accomplished when digital transformation is upgraded to a tactical level [47]. Congruent with this, we measure the two aspects of digital transformation (digital strategy and digital capability) as a unified construct [2].

Digital strategy relates to the extent to which the enterprise integrates digital technology, its business procedures as well as its products [2]. Proksch et al. [48] state digital tactics as the means by which firms strategically employ digital resources to generate unique value, representing the extension of businesses that combine digitalization into their tactical planning. While digitization can improve sustainable performance [2], several

uncertainties exist regarding the outcomes of digital transformation, potentially leading to united outcomes. For instance, it has been argued that it could lead to substantial financial burdens [21] and promote opportunities for behavior in a situation that causes a disparity in the digital transformation of the organization and its distribution channel [49]. Based on this, there are concerns among firms regarding their operations to obtain the advantages of digitalization [50]. Hence, more research is essential in the literature to demonstrate the benefits of digital transformation, particularly for sustained growth.

2.3. Green Knowledge Acquisition

To address environmental concerns, firms must acquire solution-driven information, such as green technologies and green information regarding environmental requirements and green technologies [51]. Green knowledge acquisition prioritizes green information through digital technology to impact green knowledge management and green technology innovation initiatives [11,52]. Green knowledge acquisition describes the enterprise capability that creates a green business perception from different resources that can be used by the organizational members for strategic decision-making [11,53,54]. Despite its importance, ambiguity continues to cloud the technology convergence including the firms struggling to understand their capability to obtain business values from the adoption [12]. From the perspective of new institution theory, the concept of green knowledge acquisition capabilities can broaden the scope of digital transformation to include all relevant networks that are crucial for realizing the complete tactical capability of green information. Due to limited studies on obtaining green knowledge [52], our study examines green knowledge acquisition as an appliance through which firms can use digital transformation to achieve sustainable corporate performance.

2.4. Digital Transformation and Sustainable Corporate Performance

Digital technologies like IoT (internet of things), AI (artificial intelligence), and blockchain can significantly contribute to sustainable practices in the manufacturing sector by enabling optimized resource management, improved production efficiency, and transparent supply chains [39,55]. This minimizes environmental impact and promotes responsible production methods. Gaining sustainable corporate performance is critical for all firms. The triple-bottom-line concept (i.e., economic, social, and the environment) underlines that it is crucial to integrate digital transformation [2]. From this standpoint, firms have the responsibility to ensure that their operation gives importance to the safety and health of society. According to some research [56], a positive relationship between digital transformation and sustainable performance using PLS-SEM was proposed. Based on this, digital transformation is crucial in developing a new sustainable economy [56].

The new institutional theory proposes that for a firm to make significant progress toward transformative change, it needs a robust organizational capability, which requires leveraging resources and competencies. The digital capability allows firms to employ innovative approaches and tools to address environmental and social issues. For example, digital capacity assists businesses in fulfilling their communal obligations [57] through systems and data places that can properly recognize external opportunities and streamline information flow [2]. Thus, upgrading digital infrastructure and digital technology proficiency is not only beneficial for transforming operational processes to improve resource use, it also minimizes waste, and improves information flow between diverse functions. Digital technologies can help achieve the three pillars of sustainability, environmental, social, and economic, by reducing waste, improving resource management, and promoting social inclusion [58]. The previous study found that a framework for research with digital technologies as a precursor and driver of sustainability is achievable. In the digital

revolution, the report established (i) increasing digital skills and (ii) balancing their economic, (iii) environmental, and (iv) social implications were vital factors [59]. In the circular economy context, Aboalhool et al. [60] reported that adopting digital capacity facilitates the execution of circular economy practices, including environmental management, green design, and investment recovery.

According to theory, digital transformation can significantly influence a company's sustainable corporate performance by aligning its practices with the evolving norms and expectations of stakeholders. This is particularly relevant regarding environmental and social responsibility, thereby creating a legitimate and sustainable business model through technological advancements and data-driven insights. In a previous study, digital transformation enhances the promotion of companies' positive environmental performance, and improves resource efficiency and sustainability [61]. From the new institutional perspective, traditional practices and beliefs have faded due to new practices and beliefs. Along this line, Xu et al. [2] suggested that it helps in devising digitalization direction and promoting the integration of digital infrastructure for environmental preservation initiatives, such as developing internal environmentally friendly information systems and embracing environmentally friendly design technology [2]. Further, digitalized strategy has been penetrating organizational structures in recent years as part of a new wave of digitization goals to promote the sustained performance of enterprises [23]. Finally, integrating digital transformation and sustainability within an organization presents several challenges, including cultural resistance to change, inadequate technological capabilities, data integration complexities, financial constraints, and a lack of clear metrics to measure the impact of sustainability initiatives within a digital framework [27]. Given in the above argument, this research posits the following:

H1. *Digital transformation has a positive impact on sustainable corporate performance.*

2.5. Digital Transformation and Green Knowledge Acquisition

Given the increasing pressure on manufacturing firms from various stakeholders to adopt sustainable business practices, it is critical for green competence to be achieved. Green competence describes the knowledge, skills, and abilities needed to implement environmentally friendly strategies within a company. It ultimately aims to minimize a company's ecological footprint [8,60]. To this end, Sahoo et. al. [11] argued that knowledge acquisition could be a knowledge generator for firms to solve environmental issues. Similarly, it has been argued that green knowledge is a crucial asset for firms to address environmental concerns through building process innovation and products [12]. The digital transformation overcomes green information assimilation barriers through enhanced data accessibility, automation, and collaborative tools. Advanced technologies such as cloud computing, data analytics, and artificial intelligence streamline information sharing and collaboration. It reduces costs by minimizing reliance on physical resources and traditional infrastructure, allowing organizations to access and disseminate sustainable practices quickly and efficiently.

Further, through digital transformation, firms may, to a large extent, eliminate green information assimilation barriers to assist in obtaining green knowledge at reduced cost, and facilitate the formation of knowledge spillover [5]. Through the literature review, the exact relation between digital transformation (i.e., digital strategy and digital capability) and green knowledge acquisition has not been explored. On comparing green knowledge and green competence, green knowledge refers to the theoretical understanding of environmental issues and sustainable practices, whereas green competence signifies the ability to apply that knowledge in practical situations, effectively implementing environmentally friendly strategies within a business setting. Moreover, green knowledge is information,

while green competence is the skill of using that information to make sustainable decisions and actions.

Green knowledge acquisition reflects the capability of the firm to use all its useful resources together with technical resources, to detect, manage, and establish green knowledge. Prior studies [62,63] outlined the operating part of digital capability in the creation of green knowledge: (1) digital capabilities enhance the formation of green knowledge through collaborative green R&D efforts between firms, (2) green R&D allows organizations to share knowledge and ideas on low-carbon challenges, and (3) digital capability promotes green knowledge between firms. Further, digital strategies enable firms to use digital technologies to establish an atmosphere for all organizational members to pursue the latest knowledge actively [48], which promotes the ability to detect, screen, and leverage external information [2]. However, these are theoretical explanations that need empirical investigation. Given the above, we propose the following:

H2. *Digital transformation has a positive impact on green knowledge acquisition.*

2.6. Green Knowledge Acquisition on Sustainable Corporate Performance

“Acquisition” symbolizes the capability to identify and obtain the latest knowledge, necessary for efficient firm processes [64]. Knowledge acquisition describes the procedure of obtaining and accumulating further insights from different reliable sources that will help employees address challenges related to business operations, improving employees and overall firm efficiency [13]. Such acquired knowledge is crucial in the knowledge application process and in enhancing firm-related outcomes. Firms integrate this green information into their decision-making processes by actively collecting and analyzing environmental data, incorporating sustainability metrics into key performance indicators, using specialized software tools, and developing internal knowledge management frameworks that prioritize green practices [65]. Green knowledge management is a process that integrates environmentally-friendly practices into an organization’s knowledge management system. It can help companies achieve a sustainable advantage. Guided by green information derived from green knowledge acquisition, corporate intelligence for the development of eco-processes and eco-products can be established through the co-creation of green knowledge between a variety of functional units of a firm [11,66].

Despite this, there is a scarcity of empirical studies in the manufacturing sector exploring the exact relationship between green knowledge acquisition and sustainable corporate performance [2,11]. Although previous studies examined the relationship between supplier relationship management, knowledge acquisition, and firm performance and demonstrated that knowledge acquisition has a significant impact on firm performance [11]. Similarly, Al Koliby et. al. [13] found that knowledge acquisition has a significant positive impact on sustainable performance [67]. In advancing the current literature, this study contends that green knowledge acquisition is an enabler of sustainable corporate performance. Thus, we propose the following:

H3. *Green knowledge acquisition has a positive impact on sustainable corporate performance.*

2.7. Digital Transformation and Innovation Performance

Innovation is often described as the effective implementation of new ideas that arise from the collaboration of various resources within a firm. Such a combination of diverse resources leads to the formation of the latest processes, products, or services, which the firm uses in the market to differentiate itself [68]. In addition, this combination of resources is even more crucial if it pays off and acts as a catalyst for innovation performance [69]. However, every firm, whether internal or external, has its own set of boundaries. Due

to technological advancements, the boundaries within and between firms are becoming less rigid [70]. According to Wan et al. [23], digital transformation and innovation performance exhibit a positive based on reliability testing. The new institutional theory aims to transform firms by integrating digitally-powered organizational arrangements within and between firms. Simply put, the extensive integration of digital technologies promotes information flow within and between firms, eliminating inter-firm boundaries [71]. From the new institutional theory, through digital transformation, digital technology can assist organizations in overcoming capability and resource deficiencies and use external capabilities and resources to establish the strategies required for creating new products and entering new markets. Furthermore, it has been reported that digital technologies such as blockchain technology improve collaborative innovation performance [72]. Based on this argument, we propose the following:

H4. *Digital transformation has a positive impact on innovation performance.*

2.8. Innovation Performance and Sustainable Corporate Performance

Investigating the influence of innovation performance is environmentally conscious and is essential for advancing SDG 9 [73]. From this sense, the authors of Ahmad et al. [74] identified that the capacity of a firm to innovate has a positive impact on its chances of survival and longevity. Further, other research Chawla et al. [75] demonstrates the essential functions of process and product innovation and administrative innovation on sustainability. Advancing innovation processes to adhere to stringent environmental regulations can promote sustainability by curbing pollution, minimizing energy usage, and boosting efficiency [76]. Eco-product innovation demands that firms integrate sustainability into their new product development processes by creating pollution-free and recyclable products to avoid or minimize environmental risk [77], improving sustainable performance. Based on the available evidence, we propose the following:

H5. *Innovation performance has a positive impact on sustainable corporate performance.*

2.9. Green Knowledge Acquisition as a Mediator

Although this study anticipates the advantages of the relation between sustainable performance and digital transformation, at the same time, green knowledge acquisition may affect the use of digital transformation to achieve sustainable corporate performance. Green knowledge acquisition implies the knowledge associated with an organization's environmental protection [78]. Accordingly, Pereira and Bamel [79] and Tu and Wu [80] define knowledge as a crucial strategic source for generating the latest values and a rivalrous edge. From this standpoint, firms protect their crucial resources using green knowledge, an enterprise's critical resource to accomplish sustainable corporate performance.

In addition, firms must acquire and disseminate knowledge to accomplish sustainable performance and adhere to environmental requirements [8]. As a crucial strategic resource, to obtain the advantage effectively, knowledge should be managed and gained properly. We propose that digital transformation indirectly influences sustainable corporate performance through green knowledge acquisition. Simply put, a manufacturer's sustainable performance position may rely on its environmental knowledge alongside different environmental technological capabilities to achieve a sustainable competitive performance. From the new institutional theory perspective, as firms strive to accomplish value commitments and adapt to changing conditions, firms with appropriate digital strategies must prioritize applying and advancing knowledge structures [81,82].

Digital capability offers new opportunities to acquire new knowledge from digital transformation [8]. In comparing the relationship between digital capability and digital

strategy, digital capability is a firm's ability to integrate technology and organizational resources to respond to business demands, while digital strategy is a plan for how a firm will use digital technology to achieve its goals [26]. Digital capability and digital strategy are complementary, and both are important for a firm's digital transformation. They both significantly support green knowledge acquisition by enabling efficient data collection, analysis, and sharing across an organization, facilitating collaboration with external partners, identifying sustainable practices, and ultimately driving innovation in environmentally friendly products and processes within a company. Only with digital capability and digital strategy through digital transformation can a firm efficiently discover green knowledge, determine the most appropriate methods to obtain the knowledge, and successfully integrate the green knowledge to improve efficiently, thereby moving the firm towards sustainable corporate performance. Based on the discussion above, we propose the following:

H6. *Digital transformation has a positive indirect effect on sustainable corporate performance through green knowledge acquisition, such that digital transformation positively influences green knowledge acquisition, which in turn influences sustainable corporate performance.*

2.10. Innovation Performance as a Mediator

Innovation relates to notable and leading technological drivers that have shaped the manufacturing industry over the past decades. Accordingly, the manufacturing industry has begun to expand its horizons and adapt its business models [83]. In the current era of digitalization, firms seek to use digital transformation not only to promote the effectiveness of the organization but also to integrate digital technologies to improve performance and meet consumers' environmental and social demands [84]. However, ordinary improvements in digital capability might indirectly impede sustainable development because of the competitors' ever-increasing focus and improvement on digital transformation; the performance gained through reliance on digital technologies may soon be surpassed by newcomers [5]. Innovation performance invests critical resources toward ensuring a firm's sustainability [85]. On one hand, researchers demonstrated that digital transformation drives sustainable performance [2,20]. Alternatively, there exists a contradictory warning among scholars indicating the value of digitalization will diminish when innovation remains dormant for an extended period [86,87]. Despite this, the latest literature has ignored that, acting in an intermediary role of innovation performance connecting digital transformation and sustainable corporate performance.

Furthermore, it has been suggested that when firms use specific approaches, they are likely to achieve innovative performance in terms of process and product innovation [88]. Since innovation and sustainability are commonly related to each other [89], firms must be innovative concerning the use of digital strategies to establish a business model for sustainability resulting in the growth of sustainable performance. Based on the discussion above, we propose the following:

H7. *Innovation performance plays a mediation role in the relationship between digital transformation and sustainable corporate performance.*

2.11. Digital Transformation Leadership as Moderator

Although there is no clear statement of digital transformational leadership in the current literature, digital leadership ability is linked with the leadership capability to guide a firm toward digital transformation to improve its adaptability in an ever-changing digital and social ecosystem [90]. Since digital transformation in the production industries is still a new phenomenon, and its resulting impact on sustainable development is still under

debate in the current literature [5], the conditions under which the relationship is more or less pronounced have not been established. In a scenario where the link connecting the second construct is not flexible, a third construct may influence such a relationship either by enhancing or by weakening it and, in such cases, contrary to the direction of such a relationship. Despite that, digital transformation involves different consequences that reshape business models and influence knowledge, innovation, employees, leaders, and organizational cultures and performance [2,26,63,81,91], the function of digital transformation leadership as an organizational or managerial condition has been largely understudied. Moreover, in the view of new institutional theory, leadership is integral to a firm's belief and value systems: as a firm changes, leadership must adapt and evolve [92]. As firms move toward digital transformation that fundamentally alters the operations to enhance their outputs to conform with ecological regulations and to accomplish sustainable corporate performance [2,8], leadership that helps firms bring together stakeholders and create a venue for action is required for this to occur. Hence, digital transformation leadership may implement changes in how digitalization is implemented, as a low or high impact of digital transformational leadership on digital transformation may have varying effects on green knowledge acquisition, innovation performance, and sustainable corporate performance [26].

Despite the numerous benefits that digital transformation offers firms in the modern business landscape, practitioners and academics find it difficult to acknowledge it, primarily so it differentiates the firmly established IT-powered organizational transformation [93]. Moreover, the primary determinant of success in digital transformation is the formulation of digital tactics that are compatible with business and corporate strategies [94]. Based on this, digital leadership capability assists firms in implementing changes and ensuring that no stakeholder is excluded in a digital transformation process [27]. Similarly, the execution of digital transformation demands a different organizational framework, which indicates the performance evaluation metrics [11]. While digital transformation promotes sustainable performance [2], this link may be more conducive under digital transformation leadership. For example, these leaders are efficient in empowering subordinates to adapt the essential skills in digitalization and in training them in how to utilize them effectively to drive sustainable performance [5,18].

Digital capability contributes to corporate-level innovation and innovation performance activities [71,72]. From the new institutional theory perspective, transformational leaders motivate employees to go above and beyond to seek solutions for their immediate needs and perform even better in a challenging and complicated environment [11], by boosting subordinates' creativity and encouraging them to build abilities and knowledge that drives innovation performance. An organization can foster the most innovation when its leadership style, surrounding environment, and internal structure all actively promote ambidexterity. This refers to the ability to simultaneously pursue both new ideas and optimize existing operations. Essentially, balancing creativity with efficiency to drive innovation across all levels [95]. Leadership can drive the adoption of digital technologies for sustainability by setting a clear vision for environmental impact, strategically allocating resources towards green tech, actively communicating the value of sustainability initiatives, empowering employees to participate, and fostering collaboration across departments [96]. It is essential for achieving a sustainable competitive advantage as it ensures organizations are comfortable with technological change and proactive innovation [97].

Digital leadership is essential for achieving sustainable competitive advantage as it ensures that organizations are not only responsive to immediate technological changes but also proactive innovation. It plays a crucial role in integrating advanced technologies such as AI, machine learning, and data analytics into business operations. This integration

optimizes processes, enhances productivity, and creates unique value propositions that competitors cannot easily replicate. Taken together, it is reasonable to infer that among manufacturing firms with high (low) digital transformational leadership, the impact of digital transformation on innovative green knowledge acquisition, innovation performance, and corporate sustainable performance is likely to be further strengthened (weakened), as demonstrated in Figure 1. Hence, the following are proposed:

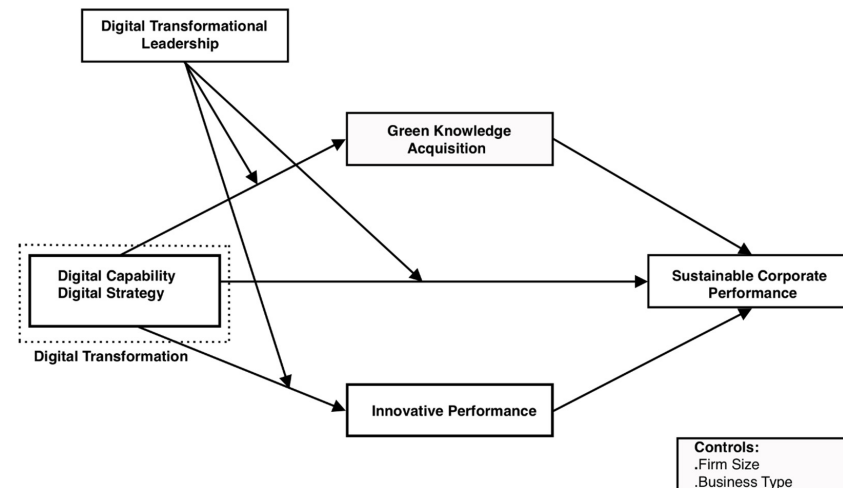


Figure 1. Integrated theoretical framework.

H8. *The positive link between digital transformation and green knowledge acquisition is further strengthened in firms with high digital transformational leadership.*

H9. *At the lower level of digital transformational leadership, the positive link between digital transformation and innovation performance diminishes.*

H10. *The positive link between digital transformation and sustainable corporate performance is further strengthened at higher levels of digital transformational leadership.*

3. Research Methodology

3.1. Research Context

We focus on the Turkish manufacturing industry for several reasons. The government plans to invest a significant amount of funds, between TRY 26 billion and TRY 39 billion (approximately USD 1–1.5 billion), to implement digital transformation solutions in the manufacturing sector. This investment aims to improve the manufacturing process by embracing digitalization.

This sector provides an intriguing research context for our study due to the current emphasis on sustainable manufacturing processes and activities, the integration of technologies into firms' processes, and its significance as a key manufacturing and distribution hub.

To analyze the impact of digital transformation on sustainability and other production activity factors, this study employs multiple statistical techniques. Descriptive statistics were used to summarize sample characteristics. To assess potential common method bias (CMB), Harman's single-factor test and the unmeasured latent method factor model were conducted in AMOS. Confirmatory factor analysis (CFA) was performed to validate measurement constructs, ensuring reliability and discriminant validity. Regression analysis was used to examine the direct effects of digital transformation on sustainable corporate performance, green knowledge acquisition, and innovation performance. Mediation effects of green knowledge acquisition and innovation performance were tested using Hayes'

PROCESS macro (Model 4) with bootstrapping (5000 samples). Moderation effects of digital transformational leadership were analyzed using Hayes' PROCESS macro (Model 8). The model's overall goodness-of-fit was evaluated using indices such as CMIN/DF, TLI, NFI, IFI, RFI, CFI, GFI, AGFI, and RMSEA to ensure a robust methodological framework.

3.2. Sampling and Data Collection

This study collected data from Turkish manufacturing firms in Istanbul, Bursa, and Izmir. These industrial cities are where most of the manufacturing firms are situated. Primary data (i.e., questionnaire survey) were collected from middle and senior-level managers of the surveyed firms through a quantitative research approach [98]. Before data collection, three professors of environmental technology and information systems reviewed the survey questions to ensure their clarity [99]. Through the purposive sampling method, data were collected from manufacturing firms listed in the Trade Gazette of Turkey through electronic surveys and in-person visits. Before data collection, firms' willingness was sought through phone calls and email to ensure that the targeted firms would participate and complete the surveys. A follow-up phone call was placed as a reminder for late respondents and to encourage them to complete the survey [100]. The surveyed firms were in food processing and manufacturing, building material manufacturing, automotive and spare parts, textile and apparel manufacturing, mechanical, metal and engineering, and medical.

A total of 512 questionnaires were sent out, and 332 responses were received, but after evaluating the responses, 15 were removed due to incompleteness, resulting in a 64.84 response rate. The sampled firms' characteristics are tabulated in Table 1. In terms of nature of business, food processing and manufacturing, 59 (18.61%), building materials manufacturing, 41 (12.93%), automotive and spare parts, 22 (6.94%), textile and apparel, 76 (23.98%), mechanical, metal, and engineering, 56 (17.67%), medical/medicine manufacturing, 39 (12.30%) and others, 24 (7.57%). In terms of firm size, the majority of the firms sampled had more than 50 employees: 247 (77.92%). In terms of age of the sampled firms, the majority were firms that have been in operation for at least 11 years, 288 (84.54%).

Table 1. Sample characteristics.

Characteristics	Classification	Frequency	Proportion (%)
Nature of business	Food processing and manufacturing	59	18.61
	Building materials manufacturing	41	12.93
	Automotive and spare parts	22	6.94
	Textile and apparel	76	23.98
	Mechanical, metal, and engineering	56	17.67
	Medical/medicine manufacturing	39	12.30
	Others	24	7.57
Firm size (number of employees)	Below 50	70	22.08
	51 to 100	181	57.10
	Above 100	66	20.82
Age of firm (years)	Below 10	49	15.46
	11 to 19	155	48.90
	20 and above	113	35.64

3.3. Measures

We used a translation and back-translation to translate the questionnaires from English to Turkish [101], as all measures were originally written in English. Digital transformation was measured with 7 items adopted from other research [48,102]. The items measured the extent to which firms prioritize digitalization in their strategies and their ability to detect, acquire, and use digital technologies. Green knowledge acquisition was measured with 6 items adopted [52]. The items reflect the extent to which firms acquire commercial and/or technical knowledge about environmental regulation and protection, environmental impacts of materials, products, and services, and environmental management techniques and manuals. Innovative performance was measured with 4 items adapted from other research [103]. The items were adapted and modified to fit the current study context from the previous organization innovation scale. The items rate the extent to which firms have made strong advances in implementing digital solutions to improve several organizational aspects.

Digital transformational leadership was measured with 4 items. The items were originally from the transformational leadership scale [104,105]. The items were then adapted and modified by AlNuami et. al. [34]. Sustainable corporate performance was measured with 6 items from other research [106–110]. The items reflect the level of development of firms in sustainability. See Appendix A, Table A1 for measurement items.

4. Results

4.1. Common Method Bias (CMB)

To minimize CMB, we adopted commonly used procedural and statistical remedies in cross-sectional research. First, different rating scales were used for the dependent variable (i.e., sustainable corporate performance) compared to other constructs [111]. Second, the participants of the survey were informed that complete anonymity would be ensured to encourage honest and sincere answers [112]. The goal of this approach was to prevent the fear of being judged and discourage them from providing appropriate answers or socially desirable answers. Third, using exploratory factor analysis: Harman's single factor was employed to check for CMB issues. Out of the total variance, the first factor only accounted for 20.55% of the variation, indicating CMB issues are not a serious problem in the present study [111]. Fourth, we employed the unmeasured latent method factor model [113], including the latent factor based on the integrated theoretical model's five-factor model. We then compared the unmeasured latent factor model (CMIN/DF = 1.387; TLI = 0.972; NFI = 0.930; IFI = 0.976; RFI = 0.919; CFI = 0.976; GFI = 0.907; AGFI = 0.887; RMSEA = 0.039) and the integrated theoretical five-factor factor model (CMIN/DF:1.544; TLI = 0.969; NFI = 0.926; IFI = 0.972; RFI = 0.916; CFI = 0.972 GFI = 0.904; AGFI = 0.883; RMSEA = 0.041). When the latent method factor was included in the model, it was found that CFI increased by 0.004 (less than 0.05), indicating the latent method factor failed to substantially improve the model, suggesting that CMB is not a serious issue in this study [113].

Fifth, following Whitney and Lindell's [114], which was recently adopted by similar research [8], we included a theoretically unrelated construct at the end of the survey with the main variables of the study. The unrelated marker variable had a correlation of less than 0.04 with the main study's construct. Based on the results of procedural and statistical remedies, CMB is not a major issue in this study.

4.2. Reliability and Validity Test

Using confirmatory factor analysis (CFA) in AMOS 24, the reliability of the constructs was evaluated through internal consistency assessment. Cronbach's alpha (α) and com-

posite reliability (CR) were used to estimate the reliability of the constructs. The results of internal consistency reliability and convergent validity are reported in Table 2. The items and the constructs satisfy the necessary measurement criteria. Specifically, all standardized factor loadings were above 0.6 (between 0.657 and 0.909), visualized in Figure 2, indicating the items' reliability was accomplished [115,116]. Moreover, the α values for each construct were above 0.7 (between 0.882 and 0.945). Also, the results show the average variance extracted (AVE) values were above 0.5 (between 0.580 and 0.706), which indicates that convergent validity is ensured [117]. Put simply, this study's integrated theoretical model has good internal consistency and convergent validity.

Table 2. Measurement model's summary.

Construct	Codes	SFL	Cronbach's Alpha	Composite Reliability	AVE
Digital transformation			0.942	0.944	0.706
	DT1	0.782			
	DT2	0.903			
	DT3	0.850			
	DT4	0.808			
	DT5	0.831			
	DT6	0.802			
	DT7	0.896			
Green knowledge acquisition			0.891	0.891	0.580
	GKA1	0.657			
	GKA2	0.682			
	GKA3	0.683			
	GKA4	0.816			
	GKA5	0.795			
	GKA6	0.904			
Innovative performance			0.882	0.885	0.660
	IP1	0.692			
	IP2	0.892			
	IP3	0.769			
	IP4	0.880			
Digital transformational leadership			0.886	0.887	0.662
	DTL1	0.784			
	DTL2	0.793			
	DTL3	0.824			
	DTL4	0.853			
Sustainable corporate performance			0.925	0.927	0.680
	SCP1	0.850			
	SCP2	0.827			
	SCP3	0.723			
	SCP4	0.794			

Table 2. Cont.

Construct	Codes	SFL	Cronbach's Alpha	Composite Reliability	AVE
	SCP5	0.909			
	SCP6	0.834			
Summary of model fitness: CMIN/DF: 480.205/311 = 1.544; $p < 0.000$; TLI = 0.969; NFI = 0.926; IFI = 0.972; RFI = 0.916; GFI = 0.904; CFI = 0.972; AGFI = 0.883; RMSEA = 0.041					

Note: DT= digital transformation; GKA= green knowledge acquisition; IP = innovative performance; DTL = digital transformation leadership; SCP = sustainable corporate performance; SFL= standardized factor loading.

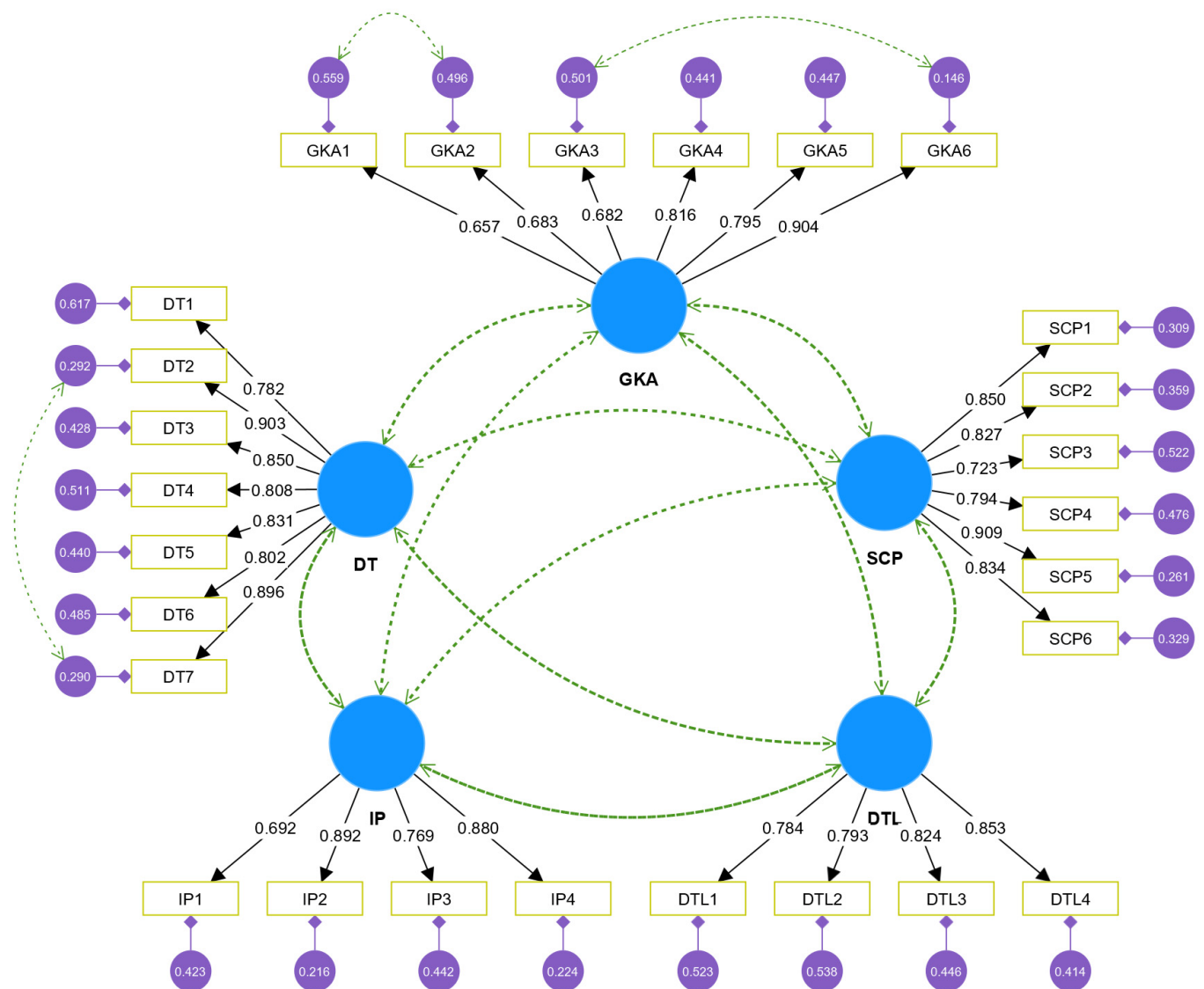


Figure 2. Standardized factor loadings.

The distinctiveness of a construct from other constructs indicates its discriminant validity. Discriminant validity was determined by comparing whether the square root of AVE values for each variable was higher than the off-diagonal correlation coefficients. Table 3 shows that the square root of all AVEs was above the off-diagonal correlation coefficients. Therefore, discriminant validity is achieved [117].

Table 3. Results of descriptive statistics, correlation, and discriminant validity (Fornell–Larcker criterion).

Construct	Mean	Standard Deviation	DT	GKA	IP	DTL	SCP	Firm Size	Business Type
DT	3.490	1.048	0.840						
GKA	3.808	0.819	0.463 **	0.761					
IP	3.654	0.793	0.579 **	0.465 **	0.812				
DTL	3.721	0.998	0.287 **	0.239 **	0.247 **	0.814			
SCP	3.838	0.918	0.397 **	0.375 **	0.401 **	0.287 **	0.824		
Firm size	1.53	0.622	0.101 **	0.036	0.021	0.029	0.019	-	
Business type	3.105	1.277	0.096	0.007	0.034	0.005	0.007	0.044	-

Note: Numbers in diagonal and bold are the square root of AVEs; ** $p < 0.01$.

Furthermore, we evaluated the model's goodness of fit, reported in Table 2. This was achieved by computing (CMIN/DF: 1.544) and (TLI: 0.969; NFI: 0.926; IFI: 0.972; RFI: 0.916), which falls between 0 and 1. However, the nearer the value to 1, the better the fit [118,119]. In addition, (GFI: 0.904; AGFI: 0.883) and (RMSEA = 0.041, which is less than 0.08), thus, the data collected fits well with the model.

4.3. Testing of Hypotheses: Regression Results for Direct and Indirect Effects

We ran Model 4 in Hayes' PROCESS macro plug-in to test the direct and indirect effects. Before the analysis, all constructs were mean-centered [98,120]. The results of the regression analysis are reported in Table 4. It was found that digital transformation has a positive effect on sustainable corporate performance (coeff. = 0.186, $t = 3.220$, $p < 0.01$). Digital transformation has a positive effect on green knowledge acquisition (coeff. = 0.368, $t = 9.562$, $p < 0.001$). Green knowledge acquisition has a positive effect on sustainable corporate performance (coeff. = 0.199, $t = 2.966$, $p < 0.01$). Digital transformation has a positive effect on innovative performance (coeff. = 0.449, $t = 13.218$, $p < 0.001$). Innovative performance has a positive effect on sustainable corporate performance (coeff. = 0.220, $t = 2.902$, $p < 0.01$). Considering these regression results, H1–H5 were supported.

Table 4. Direct effects and mediation analyses (indirect effects).

Bootstrapping results for direct (bias-corrected CI technique 95%)						
	Path	Coeff	SE	t	LLCI	ULCI
H1	DT → SCP	0.185	0.057	3.220 **	0.072	0.299
H2	DT → GKA	0.368	0.038	9.562 ***	0.382	0.516
H3	GKA → SCP	0.199	0.067	2.966 **	0.067	0.331
H4	DT → IP	0.449	0.034	13.218 ***	0.382	0.516
H5	IP → SCP	0.220	0.076	2.902 **	0.071	0.370
Indirect effects		Indirect using the bootstrapping method				
	Effect	BootSE	BootLLCI	BootULCI		
Total		0.172	0.044		0.090	0.264
H6	DT → GKA → SCP	0.073	0.034		0.008	0.142
H7	DT → GKA → SCP	0.099	0.038		0.029	0.177

Note: Unstandardized regression coefficients were reported; bootstrap resample size = 5000; ** $p < 0.01$, *** $p < 0.001$; CI = confidence interval; LL = lower limit; UL = upper limit.

Following Baron and Kenny's [121] procedure for examining mediating effects, we examined the mediating roles of green knowledge acquisition and innovation performance. When green knowledge acquisition and innovation performance were added as mediators in the model, the direct effect of digital transformation remained significant. Therefore, green knowledge acquisition and innovation performance partially mediate the aforementioned relationship. To ensure the robustness of the mediating effects, we adopted the bootstrapping method (with 5000 samples) to probe the indirect effect further. According to several studies [122–124], the bootstrap method provides a thorough understanding of indirect effects. A significant indirect effect is observed when zero is not within the confidence intervals [125]. In Table 4, the indirect effects for green knowledge acquisition (coeff. = 0.073, BootLLCI = 0.008, BootULCI = 0.142) and innovative performance (coeff. = 0.099, BootLLCI = 0.029, BootULCI = 0.177) are shown. Because zero is not within the confidence intervals, the indirect effects of digital transformation on sustainable corporate performance through green knowledge acquisition and innovation performance were confirmed. Therefore, H6 and H7 were validated.

4.4. Testing of Hypotheses: Regression Results for Moderation Analyses

We ran Model 8 in Hayes' PROCESS macro plug-in to test the direct and indirect effects [126]. Similar to mediation analyses, all constructs were mean centered [120] and firm size and business type were included in the model as covariates. The results of the conditional effects are reported in Table 5. Table 5 reports that in Model 1, digital transformation has a positive effect on green knowledge acquisition (coeff. = 0.361, $t = 3.005$, $p < 0.01$), and the interaction term between digital transformation and digital transformational leadership moderates this relationship (coeff. = 0.149, $t = 3.815$, $p < 0.001$). We computed a simple slope test to explore (± 1 SD from the mean) digital transformational leadership to plot the interaction effect (Aiken and West, 1991). This approach is important to gain more insight into the interaction effects. The graphical representation of the interaction plot is presented in Figure 3; at high digital transformational leadership (i.e., $+1$ SD), digital transformation had a stronger significant positive effect on green knowledge acquisition (coeff. = 0.516, $t = 9.579$, $p < 0.001$, CI [0.402, 0.629]), whereas at low digital transformational leadership (i.e., -1 SD), the relationship was weaker (coeff. = 0.207, $t = 3.935$, $p < 0.001$, CI [0.103, 0.310]); this supports H8.

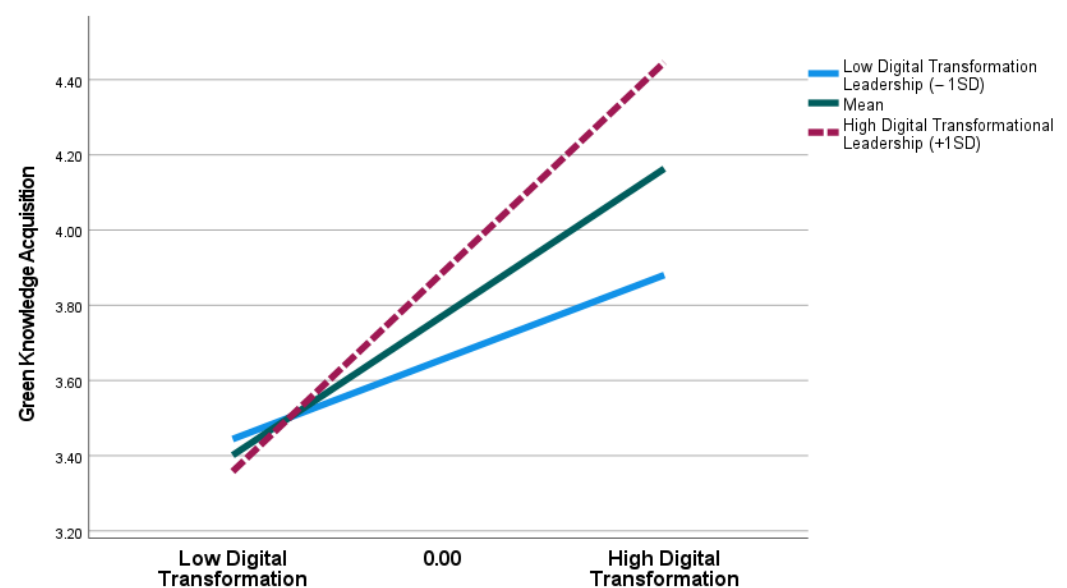


Figure 3. Shows different levels of digital transformational leadership (± 1 SD from the mean) on the influence of digital transformation on green knowledge acquisition.

Table 5. Test of a moderating effect of digital transformational leadership (moderated mediation).

Path	Coeff.	SE	t	95% Confidence Interval	
				Lower-Level	Upper-Level
Model 1:					
Firm size → green knowledge acquisition	0.034	0.017	0.037 ns	−0.061	0.012
Business type → green knowledge acquisition	0.019	0.011	0.028 ns	−0.077	0.006
Digital transformation → green knowledge acquisition	0.361	0.037	9.579 ***	0.287	0.436
Digital transformation leadership → green knowledge acquisition	0.116	0.038	3.005 **	0.040	0.192
H8: Digital transformation×digital transformational leadership → green knowledge acquisition	0.149	0.039	3.815 ***	0.072	0.226
The specific conditional values of digital transformational leadership					
−1SD (below the mean)	0.207	0.053	3.935 ***	0.103	0.310
Mean	0.361	0.038	8.896 ***	0.287	0.436
+1SD (above the mean)	0.516	0.058	9.579 ***	0.402	0.629
Model 2:					
Firm size → innovative performance	0.041	0.020	0.049 ns	−0.051	0.009
Business type → innovative performance	0.056	0.021	0.069 ns	−0.067	0.011
Digital transformation →innovative performance	0.452	0.034	13.125 ***	0.384	0.519
Digital transformation leadership → innovative performance	0.016	0.035	0.472 ns	−0.053	0.086
H9: Digital transformation×digital transformational leadership →innovative performance	0.072	0.036	2.025 *	0.021	0.143
The specific conditional values of digital transformational leadership					
−1SD (below the mean)	0.377	0.048	7.858 ***	0.282	0.471
Mean	0.452	0.034	9.952 ***	0.384	0.519
+1SD (above the mean)	0.526	0.052	13.124 ***	0.422	0.631
Model 3:					
Firm size → sustainable corporate performance	0.039	0.0176	0.044 ns	−0.006	0.007
Business type → sustainable corporate performance	0.028	0.016	0.027 ns	−0.066	0.010
Digital transformation →sustainable corporate performance	0.174	0.057	3.038 **	0.061	0.288
Green knowledge acquisition → sustainable corporate performance	0.147	0.067	2.171 *	0.013	0.281
Innovative performance → sustainable corporate performance	0.222	0.074	2.986 **	0.075	0.368
Digital transformation leadership → innovative performance	0.188	0.045	4.102 ***	0.097	0.278

Table 5. Cont.

Path	Coeff.	SE	t	95% Confidence Interval	
				Lower-Level	Upper-Level
H10: Digital transformation $\times$ digital transformational leadership $\rightarrow$ sustainable corporate performance	0.006	0.047	0.015 ns	−0.092	0.093
Index of moderated mediation by digital transformational leadership					
	Index	BootSE		BootLLCI	BootULCI
Digital transformation $\rightarrow$ green knowledge acquisition $\rightarrow$ sustainable corporate performance	0.022	0.015		0.002	0.058
Digital transformation $\rightarrow$ innovative performance $\rightarrow$ sustainable corporate performance	0.016	0.017		−0.02	0.042

Note: Unstandardized regression coefficients were reported; bootstrap resample size = 5000; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; CI = confidence interval; LL = lower limit; UL = upper limit; ns = not significant.

Table 5 reports that in Model 2, digital transformation has a positive effect on innovative performance (coeff. = 0.452, $t = 13.125$ $p < 0.001$), and the interaction term between digital transformation and digital transformational leadership moderates this relationship (coeff. = 0.072, $t = 2.025$, $p < 0.05$). The simple slope test was plotted to gain more insight into the significant interaction term. Figure 4 shows that digital transformation had a stronger significant positive effect on innovative performance (coeff. = 0.526, $t = 13.124$, $p < 0.001$, CI [0.282, 0.471]), whereas at low digital transformational leadership (i.e., $-1SD$), the relationship was weaker (coeff. = 0.377, $t = 7.858$ $p < 0.001$, CI [0.282, 0.471]); this supports H9.

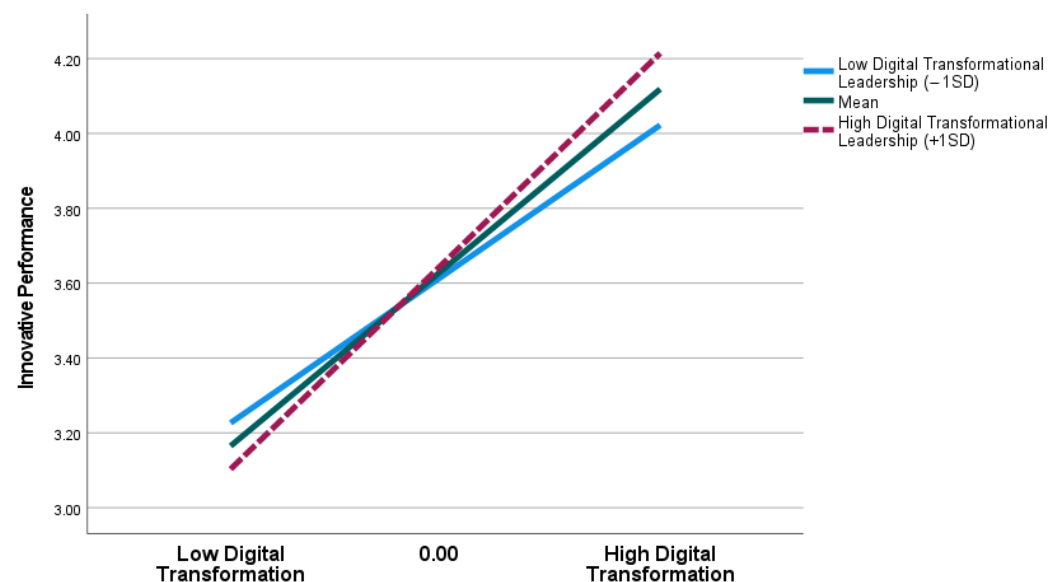


Figure 4. Shows different levels of digital transformational leadership ($\pm 1SD$ from the mean) on the impact of digital transformation on innovative performance.

In Model 3, digital transformation has a positive effect on sustainable corporate performance (coeff. = 0.174, $t = 3.038$ $p < 0.01$). However, the interaction term between digital transformation and digital transformational leadership moderates this relationship (coeff. = 0.006, $t = 0.047$, $p > 0.05$), so H10 was not validated.

5. Discussion

Drawing upon the new institution theory, this study investigated an integrated theoretical model consisting of several constructs regarding the impact of digital transformation on sustainable corporate performance and how the mediating mechanisms of green knowledge acquisition and innovation performance influence this relationship under the moderating role of digital transformation leadership in Turkish manufacturing industries.

The results indicate that digital transformation has a positive impact on sustainable corporate performance. This is enabled by improving operational efficiency, fostering innovation, and creating new business models. This also aligns with the recent research findings of Xu et al. [2] and is similar to previous research [23]. An explanation for this finding is that, from the new institutional theory perspective, digital capability promotes environmental management capability and facilitates the upgrade of operational procedures and the development of products that shift organizational change toward achieving sustainable goals. Moreover, digital strategy empowers firms to improve digitalization regularly, encourages employees to nurture digital skills, and establishes a robust sustainable management system [48]. Digital transformation has a positive impact on green knowledge acquisition. Here, the institutional pressure from regulators, customers, and competitors can also impact a company's green practices. These forces can push Turkish businesses to adopt digital technologies while simultaneously considering environmental impact. This drives towards a more sustainable digital transformation journey [127].

In Turkey, companies grew aggressively in a short amount of time, so there was no time to standardize production. In addition, even if a particular product is produced in the same type of machine in different factories, it is usually up to the workers who operate it to establish what kind of standard that product holds. To overcome this, when new products first enter the Turkish market, sometimes they are not compatible, so by implementing SAP personally, this would implement localization in Turkey. This finding provides empirical support for the argument that investment in digital transformation strengthens firms' potential to acquire green knowledge that can be used to make swift and rational decisions and further green-knowledge building [11,128]. Further, the digital strategy helps firms acquire eco-management information trends that boost rational green knowledge building. Digital transformation can transform traditional knowledge system frameworks and reshape green knowledge acquisition capabilities.

Green knowledge acquisition has a positive impact on sustainable corporate performance. It helps to reduce the environmental impact by improving resource efficiency. This finding is similar to the conclusion [11,129] that green knowledge acquired can be exploited and applied in the environmental management process to achieve sustainable targets. Obtaining green knowledge will promote the quest for sustainable performance. From this perspective, green knowledge acquisition is an intangible in a manufacturing set-up that can be used to build sustainable corporate performance. Digital transformation has a positive impact on innovation performance. This improves a company's ability to integrate internal and external resources, and by reducing the cost of an information search. And it also helps companies to manage the innovation process more effectively. This result aligns with previous research findings [23,70]. An explanation for these consistent research findings is that digital transformation encourages R&D investments [130] and enhances corporate managers' innovation awareness [70]. Through digital transformation, firms can relate and assimilate how rival firms are innovating and then be compelled to innovate. Innovation performance has a positive impact on sustainable corporate performance. This particular result strengthens previous research findings [56]. Innovation processes such as eco-process and eco-product innovation ensure that sustainability initiatives are integrated into new product development to improve sustainable performance [77].

Green knowledge acquisition and innovation performance partially mediated the digital transformation–sustainable corporate performance relationship. The previous study confirms that the perspective of innovation performance, and green knowledge acquisition can significantly improve economic efficiency in the sectors and have a positive spatial spillover effect on sustainable performance [131]. Knowledge acquisition and management are essential to successfully implement digital transformation efforts [132]. Through the intangible asset of green knowledge acquisition, digital transformation can take the more intuitive actions required in a firm to implement changes that promote sustainable development. Further, a possible explanation for the mediating role of innovation performance is that digital strategy can speed up innovation performance and direct strategic focus to digitalization to improve innovation performance, guide firms to consistently improve processes, and redefine business models to achieve sustainable corporate performance.

Digital transformational leadership moderates and further strengthens the digital transformation–green knowledge acquisition relationship in manufacturing firms with high digital transformational leadership. Based on the new institutional framework, digital transformation leaders are better equipped to handle organizational change, particularly in the digital era [81]. This finding demonstrates that high digital transformation leadership is a driving force that triggers firms to embrace digitization and its adaptation to acquire crucial green knowledge. In the previous study, the effect of digital platforms on leadership was positive and significant. Thus, it confirms that a digital platform has a direct positive effect on digital leadership. Digital training can be beneficial for improving employee performance because employees can access training resources through the internet all over the world. So, the hypothesis stating that there is a positive impact from digital platforms on leadership is accepted [26]. Furthermore, digital transformational leadership moderates and weakens the digital transformation–innovation performance relationship in manufacturing firms with low digital transformational leadership. Leadership helps organizational members become digitally proficient by providing appropriate training and motivating them to partake in digital transformational processes [81]. In contrast, with low-level digital transformation leadership, adaption to digitalization may be low, and functional units that drive innovation may lack the expertise to drive innovation performance.

6. Conclusions

6.1. Theoretical Contribution

This research offers various theoretical contributions that advance the current digital transformation and sustainability literature. First, the majority of the studies on digital transformation and digitalization in the current literature in the manufacturing context have mainly used theories such as the resource-based view, dynamic capability, organizational inertia, and information processing theory [133,134], and no theory [2,135]. This study adds to the limited body of knowledge that uses the new institutional theory [136] and demonstrates the apparent value of the new institutional theory in empirically investigating the influence of digital transformation on sustainable performance and how green knowledge acquisition and innovation performance act as mediators, and also the role of digital transformational leadership in this regard.

Second, grounded in the new institutional theory, this research provides new empirical insights that confirm the critical role of digital transformation in achieving sustainable corporate performance in the context of the Turkish manufacturing industry, extending the emerging research on digital transformation and sustainability [2]. Third, the relationship between digital transformation and green knowledge acquisition has not been previously examined in the extant literature. According to Proksch et al. [48], an internal digital

culture founded on digital strategy establishes a conducive environment for organizational members to seek new knowledge actively. This research advances the new institutional theory by conceptualizing the role of digital transformation and extracting the theoretical perspective about green knowledge acquisition. Further, this study demonstrates the crucial role of green knowledge acquisition and innovation performance in achieving sustainable corporate performance.

Fourth, congruent with the new institutional theory, organizational processes and procedures are crucial in achieving corporate-level outcomes [34,81]. However, firms encounter challenges with a strategy-to-execution gap, which leads to expensive digital transformation failure [137]. Different reasons contribute to this, including the mechanisms through which digital transformation transcends into sustainable development goals, which have not been established in the current literature. We conceptualize and offer empirical evidence that green knowledge acquisition and innovation performance are facilitating mechanisms between digital transformation and sustainable corporate performance. Thus, this research extends the scope of the new institutional theory and demonstrates that digital transformation offers an upgrade of operational changes necessary to acquire green knowledge and enhance innovation performance, moving firms toward sustainable development targets.

Fifth, the current literature acknowledges the crucial role of leadership in a digitally driven era and environmental management processes in their frameworks [34,81,136]. Despite this, the role of leadership in digital transformation is still in its early stages [26,34,81]; digital transformational leadership as an organizational condition is largely understudied. Specifically, little is known about the conditions under which digital transformation improves sustainable development; how digital transformation leadership might trigger the implementation of digital transformation on green knowledge acquisition, innovation performance, and sustainable corporate performance has not been investigated to date.

Finally, the questions in the research indicate that digital transformation substantially promotes green innovation. Green knowledge acquisition and innovation performance can mediate the relationship between digital transformational leadership and sustainable corporate performance. In previous research, the manufacturing sectors faced many difficulties in digital transformation and lacked adequate knowledge regarding technologies. It created a gap in technological updates [138]. This research has been overtaken by suggesting that digital transformational leadership can significantly strengthen the positive relationship between digital transformation, green knowledge acquisition, innovation performance, and sustainable corporate performance, acting as a catalyst for integrating sustainability practices into a company's digital strategy, thereby enhancing its overall environmental impact. This research reveals the conditions for these relationships are more or less profound. Specifically, a high level of digital transformation leadership further strengthens the impact of digital transformation on green knowledge acquisition, and a low level of digital transformation leadership weakens the impact of digital transformation on innovation performance. An important contribution to the theory underlying this research that is currently not available in the literature.

6.2. Implications for Theory

This research not only offers crucial implications for the practitioners on how digital transformation, green knowledge acquisition, and innovation performance influence sustainable corporate performance but also provides new knowledge that advances the understanding of the impact of digital transformation on green knowledge acquisition while also offering practical guidelines for manufacturing firms on how to acquire green knowledge, enhance innovation performance to accomplish sustainable performance tar-

gets by implementing digital transformation. To begin with, digitalization is a crucial way of acquiring green knowledge and achieving sustainable corporate performance. Hence, firms need to manage the processes of digital transformation through capability building and strategic formulation. This can be achieved by integrating digitalization into strategic planning processes and providing guidance to all functional units regarding the coordinated planning and implementation of digital transformation. From this standpoint, management should acquire and learn new knowledge regarding digitalization, and nurture digital cognition through interactive exchanges and regular training. Further, tools such as collaborative office systems, interconnection systems, and mobile communication can be used by firms to establish an effective three-dimensional interactive network to connect firms with external stakeholders such as suppliers and customers to create an interactive, informative chain that can promote managerial decision-making [2,34,54].

Furthermore, green knowledge acquisition is an enabler of sustainable development. Firms must thoughtfully consider how digital transformation can be successfully implemented. This involves aligning digital strategies and capabilities to facilitate green knowledge acquisition and sharing obtained green knowledge among all functions and departments to achieve sustainable corporate performance. In addition, managers should realize the importance of digital transformation and use it to foster improved innovation performance, which in turn will lead to achieving sustainable development targets.

If not appropriately adjusted, digital transformation may lead to bottlenecks [81]. Accordingly, the firms' leaders must have robust leadership capability in digitalization to ensure that digital transformation remains effective. Digital transformation is a radical organizational change that disrupts conventional methods. Thus, the findings of this study suggest that managers must possess technical capabilities in the latest digital technologies to improve green knowledge acquisition capabilities and innovation performance. In addition, firms should hire managers with high digital capabilities and green knowledge management values.

6.3. Limitations and Direction for Future Studies

While this research makes significant empirical and theoretical contributions, it is not without limitations, which offer valuable opportunities for future investigation. The restricted generalizability of the findings stems from the sample's confinement to Turkish manufacturing industries. Expanding the sample to include other sectors would enhance the robustness and general applicability of the results. Additionally, the study adopts a cross-sectional research design, which captures insights at a single point in time. Employing case studies or longitudinal research approaches in future studies would provide a deeper understanding of the progression and long-term impact of digital transformation. Future research could also explore alternative mechanisms to provide more comprehensive insights into sustainable corporate performance. Future models could incorporate constructs like green knowledge management, stakeholder integration, and shared vision to enhance the analysis. Moreover, potential contingency factors (moderators) may influence the relationship between digitalization and environmental management capabilities. To learn more about how digital transformation can help reach sustainable development goals, factors like environmental change could be added to the current framework.

Furthermore, this study does not explicitly account for the role of external institutional pressures, such as regulatory frameworks, government policies, industry standards, and competitive market forces, in shaping firms' green innovation strategies. While the research primarily focuses on internal organizational factors, external pressures often play a significant role in driving sustainability initiatives. Institutional forces can influence a firm's ability to acquire green knowledge, implement eco-innovation, and adapt digital trans-

formation strategies in response to evolving environmental expectations. Future research should consider integrating institutional pressures as moderating or mediating variables to better understand how external factors interact with internal capabilities in achieving sustainable corporate performance. Additionally, cross-industry and cross-country comparative analyses could offer further insights into how different regulatory environments impact firms' digital sustainability strategies.

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Appendix A

Table A1. Measurement items.

Digital transformation [48,102]. Please evaluate your company's performance in the following aspects, using "1" indicates "strongly disagree" and "7" indicates "strongly agree".
Digital strategy [48,102]
1. We investigate the newest trends and future scenarios in digitalization to stay competitive
2. Digital projects have a high priority within our business
3. We constantly update and refine our digital strategy
Digital capability [48,102]
1. Acquiring important digital technologies
2. Identifying new digital opportunities
3. Mastering state-of-the-art digital technologies
Green Knowledge Acquisition [52]. Each item was measured using a scale from 1 (I strongly disagree) to 7 (I strongly agree). "Thanks to international experience. . .
1. Our company acquires know-how related to environmental protection
2. Our company acquires business knowledge related to environmental protection
3. Our company acquires knowledge about the environmental effects of products, materials and services
4. Our company acquires knowledge about environmental regulation
5. Our company acquires knowledge about environmental management manuals and techniques
6. Our company acquires knowledge about environmental efficiency
Innovative Performance [103]
Each item was measured using a scale from 1 (I strongly disagree) to 7 (I strongly agree). "In the last three years our company has made strong advances in:

Table A1. Cont.

1. The implementation of digital solutions to enhance better knowledge management systems
2. The implementation of digital solutions to increase organizational flexibility
3. The implementation of digital solutions to develop stronger external relations
4. The implementation of digital solutions to obtain a higher success rate in new product launch as compared to competitors
Digital Transformational Leadership [34]
1. Our company inspires all members with the digital transformation plans for our organization
2. Our company provides a clear digital transformation vision for the organization's members to follow
3. Our company motivates team members to work together for the same digital transformation goals
4. Our company stimulates all members to think about digital transformation ideas
Sustainable Corporate Performance [106–110]
Please evaluate your company's performance in the following aspects, using "1"—"strongly disagree" and "7"—"strongly agree".
1. The efficiency of resource usage increases over time
2. There is a growth in contribution to social welfare
3. Environmental efficiency is improving over time (reducing pollution, emissions and dust)
4. There is a growth in revenue over time
5. There is a growth in profit over time
6. Growth in market-share over time
Scale: (1) "Not at all" (2) "Very low" (3) "Low" (4) "High" (5) "Very high"

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